

TCP/IP Protocol Suite, Flow Control & CIDR Subnetting

Subject: CS501 | Unit 2: Transport & Network Layers

1. Executive Overview & Syllabus Scope

Complete guide on IP addressing, CIDR, VLSM calculation, TCP 3-way handshake, TCP Congestion Control Slow Start, Conge

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 2: Computer Networks & Transport Architecture
- 1. OSI 7-Layer Model vs TCP/IP 4-Layer Architecture and protocol mapping
- 2. IPv4 Addressing & CIDR: Subnet masks, network ID, broadcast address, and VLSM formulas
- 3. Sliding Window Protocols: Stop-and-Wait, Go-Back-N GBN, and Selective Repeat SR
- 4. TCP Connection Management: 3-Way Handshake SYN, SYN-ACK, ACK and 4-Way Teardown FIN
- 5. TCP Congestion Control: Slow Start, Congestion Avoidance, Fast Retransmit, Fast Recovery
- 6. Application Protocols: DNS recursive/iterative lookups, HTTP/1.1 vs HTTP/2 multiplexing

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.